

Name : Fahad Ali

Roll no : 23P-0569

Section : BCS-4k

Lab Task 4

File Management

C programme SS:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;
    char marks[5][10];
    int total = 0;
    int mark;
    int i;
    char buffer[50];

    fd = open("marks.txt", O_CREAT | O_WRONLY, S_IRUSR | S_IWUSR);    //O_CREAT will creates the file if it does not exist.....
    if (fd == -1) {
        perror("Error creating file");
        exit(1);
    }

    for (i = 0; i < 5; i++) {
        printf("Enter marks for subject %d: ", i + 1);
        scanf("%d", &mark);
        snprintf(marks[i], sizeof(marks[i]), "%d\n", mark);
        // Write the marks to the file
        if (write(fd, marks[i], sizeof(marks[i])) == -1) {
            perror("Error writing to file");
            close(fd);
            exit(1);
        }
    }

    close(fd);    //this will close the file after writing .....

    fd = open("marks.txt", O_RDONLY);    // now it will open the file for reading marks.....
    if (fd == -1) {
        perror("Error opening file for reading");
        exit(1);
    }

    //read the marks from the file into the bufferr.....
    ssize_t bytesRead;
    int j = 0;
    while ((bytesRead = read(fd, buffer + j, 1)) > 0) {
        j++;
    }
    buffer[j] = '\0';
```

```
printf("\nMarks read from the file:\n");
for (i = 0; i < 5; i++) {

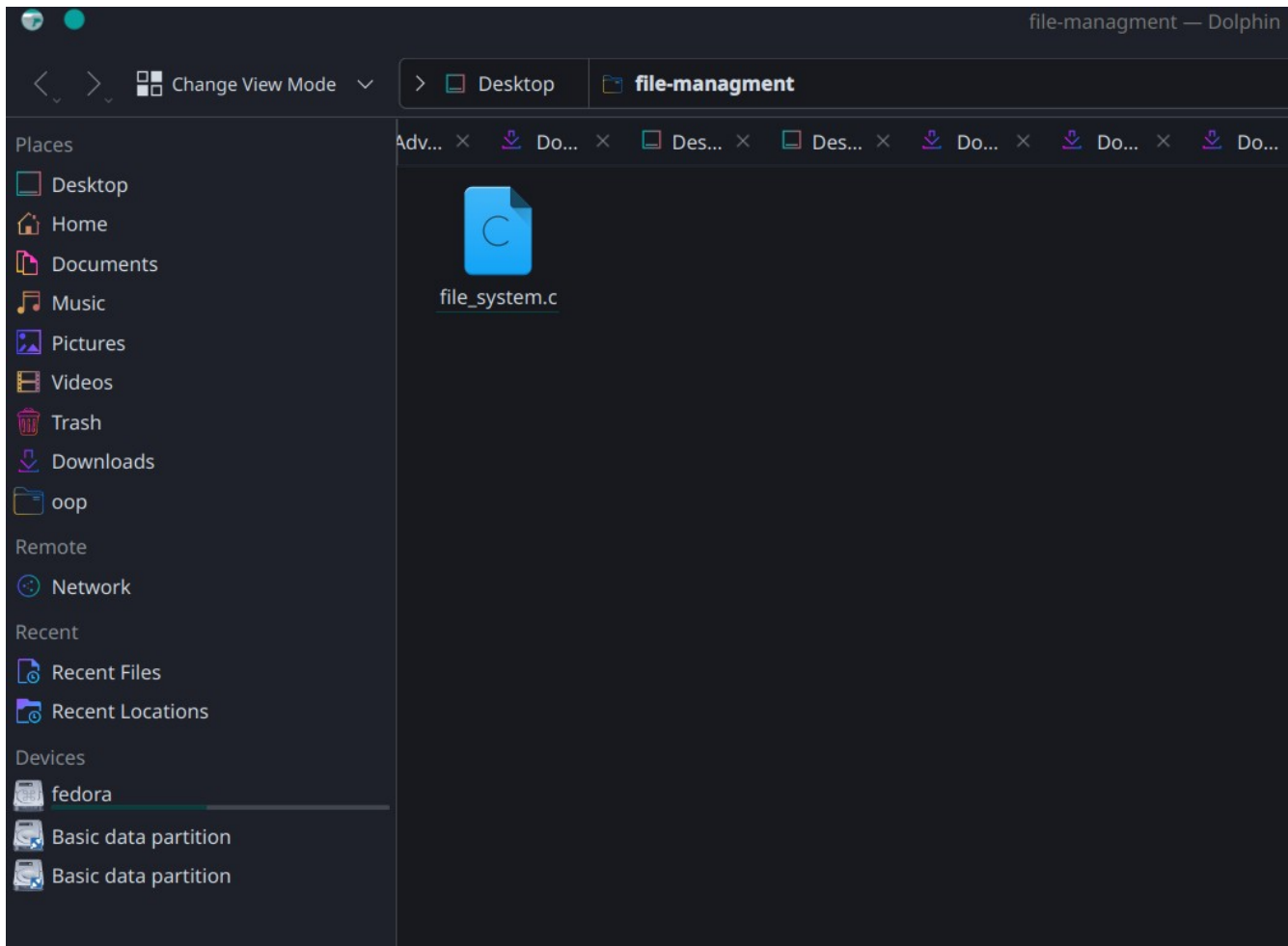
    sscanf(&buffer[i * 3], "%d", &mark);    // this will extract each mark from file.....
    total += mark;
    printf("Subject %d: %d\n", i + 1, mark);
}

printf("\nTotal Marks: %d\n", total);

close(fd);

return 0;
}
```

No mark.txt file before executing the program:



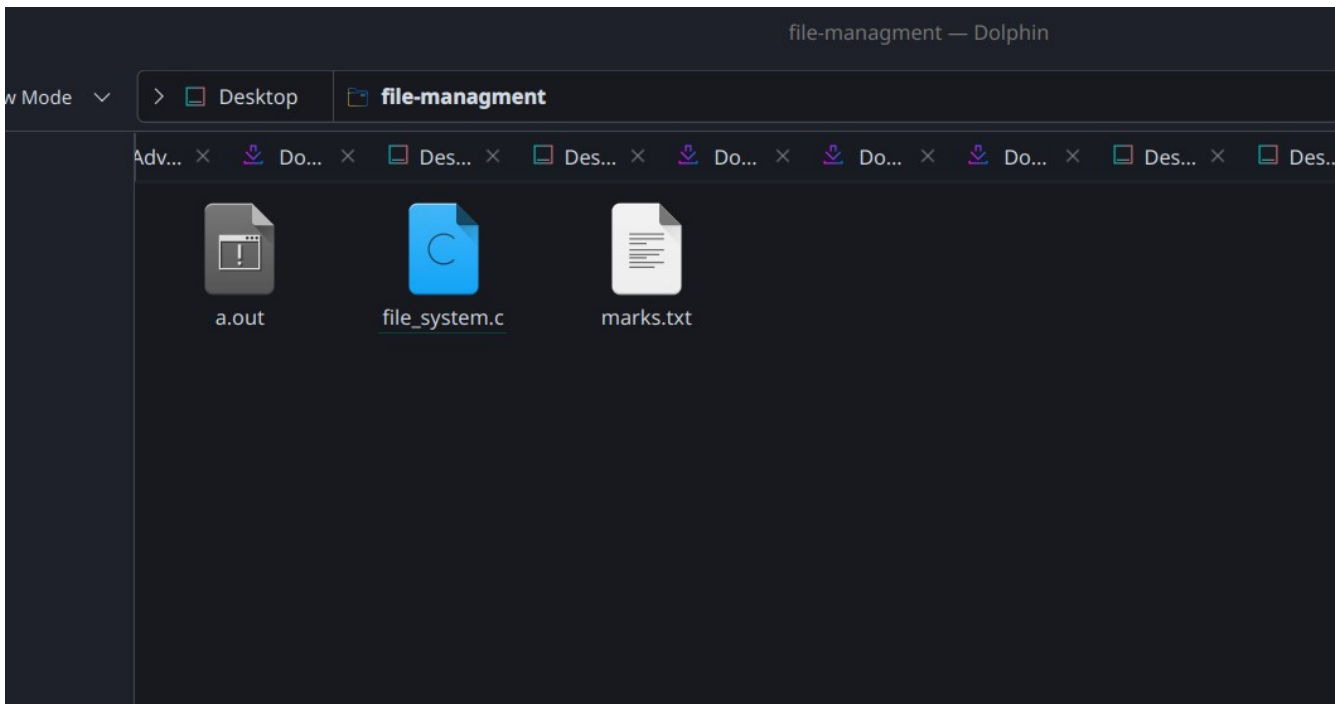
After executing the program:

```
fkhoar@fedora:~$ cd Desktop
fkhoar@fedora:~/Desktop$ cd file-managment
fkhoar@fedora:~/Desktop/file-managment$ gcc file_system.c
fkhoar@fedora:~/Desktop/file-managment$ ./a.out
Enter marks for subject 1: 50
Enter marks for subject 2: 70
Enter marks for subject 3: 80
Enter marks for subject 4: 90
Enter marks for subject 5: 75

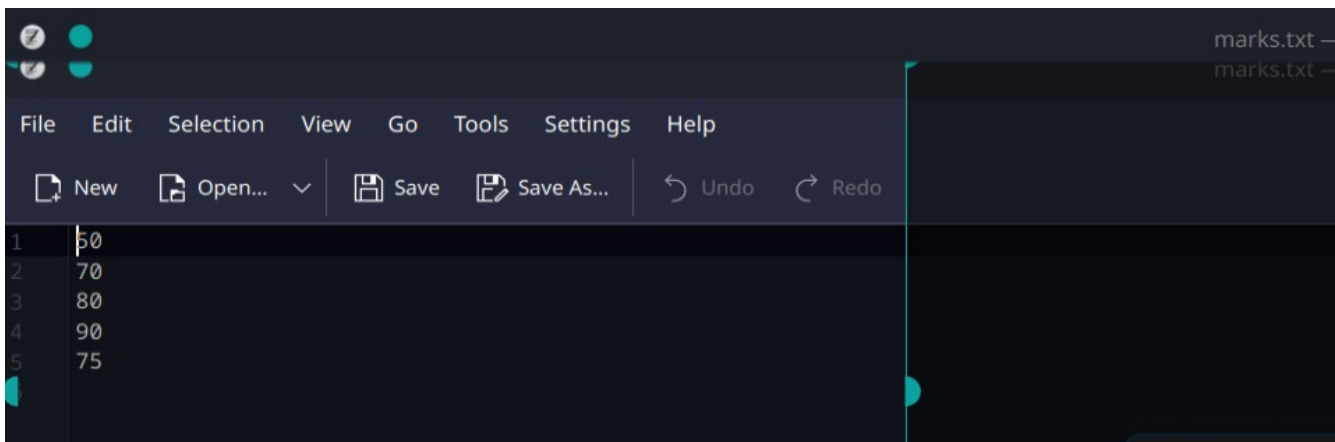
Marks read from the file:
Subject 1: 50
Subject 2: 50
Subject 3: 50
Subject 4: 50
Subject 5: 50

Total Marks: 250
```

mark.txt file created



Enterd Marks in mark.txt file:



Output:

```

New Tab Split View Copy Paste
fkhokar@fedora:~$ cd Desktop
fkhokar@fedora:~/Desktop$ cd file-managment
fkhokar@fedora:~/Desktop/file-managment$ gcc file_system.c
fkhokar@fedora:~/Desktop/file-managment$ ./a.out
Enter marks for subject 1: 50
Enter marks for subject 2: 70
Enter marks for subject 3: 80
Enter marks for subject 4: 90
Enter marks for subject 5: 75

Marks read from the file:
Subject 1: 50
Subject 2: 50
Subject 3: 50
Subject 4: 50
Subject 5: 50

Total Marks: 250
fkhokar@fedora:~/Desktop/file-managment$
```